
Détection de changements dans les séries temporelles de stations géodésiques

Kevin Gobron ^(1,2)

kevin.gobron@ign.fr

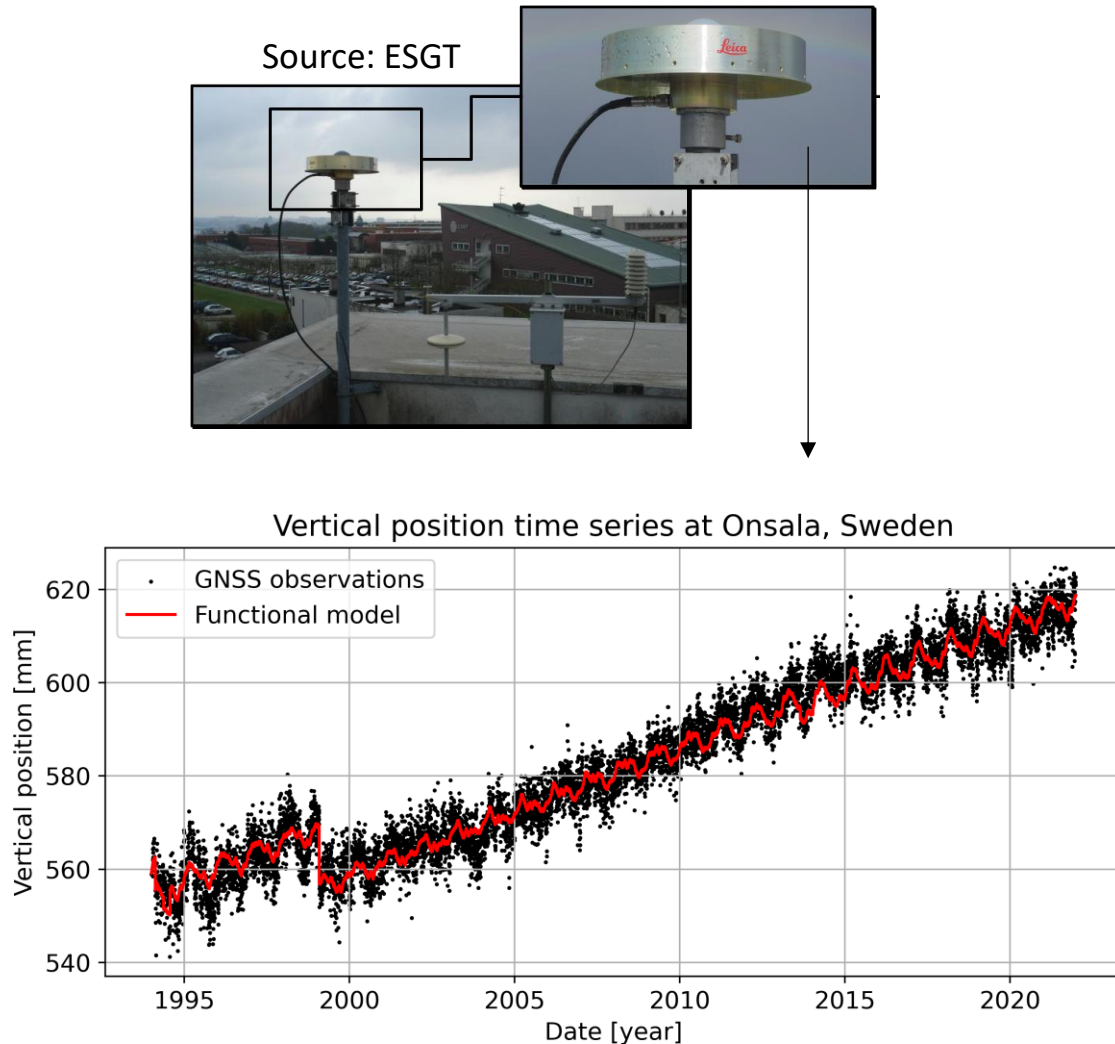
(1) - Université Paris Cité, Institut de physique du globe de Paris, CNRS, IGN, Paris, France

(2) - Ecole Nationale des Sciences Géographiques, IGN, Marne-La-Vallée, France



Example: GNSS position time series

GNSS station position time series

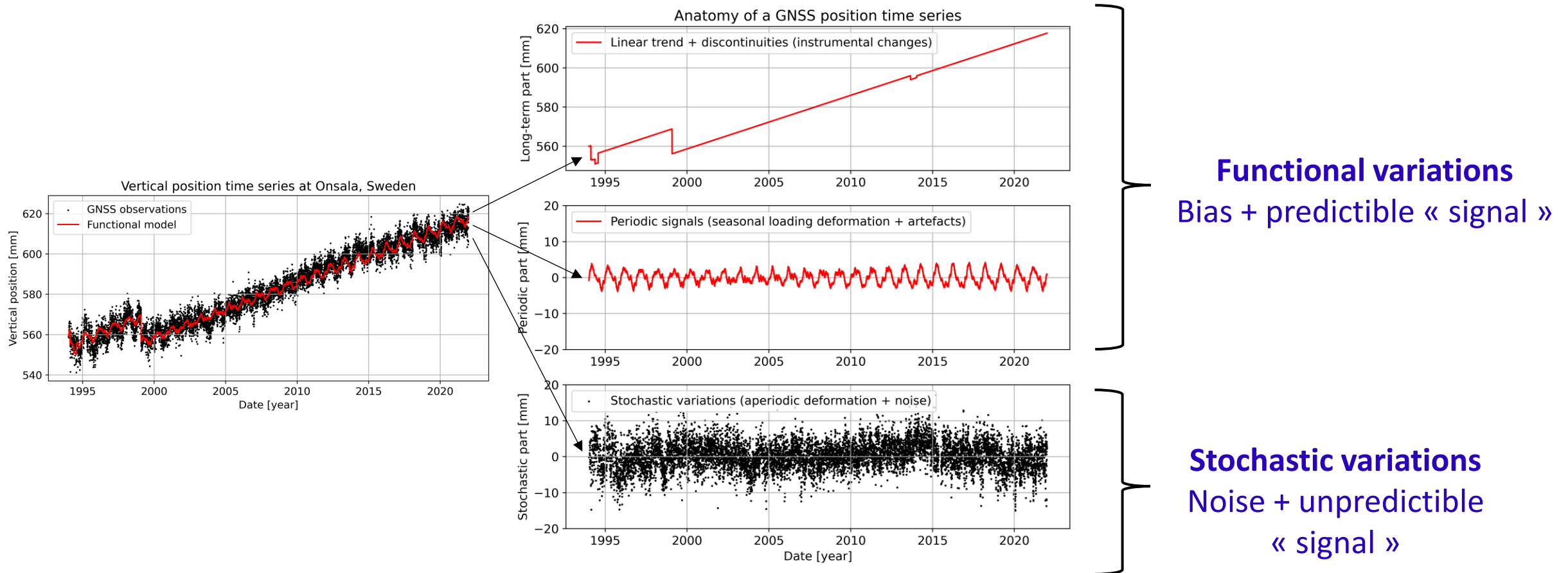


Applications

- **Terrestrial reference frame determination:** Precise positioning of ground-stations and Earth observing satellites
- **Sea-level change:** Vertical land motion correction of coastal tide-gauge records
- **Water-cycle:** Observation of the deformation induced by surface fluid mass variations
- **Ice-melting:** Observation of the deformation induced by past and present-day ice melting

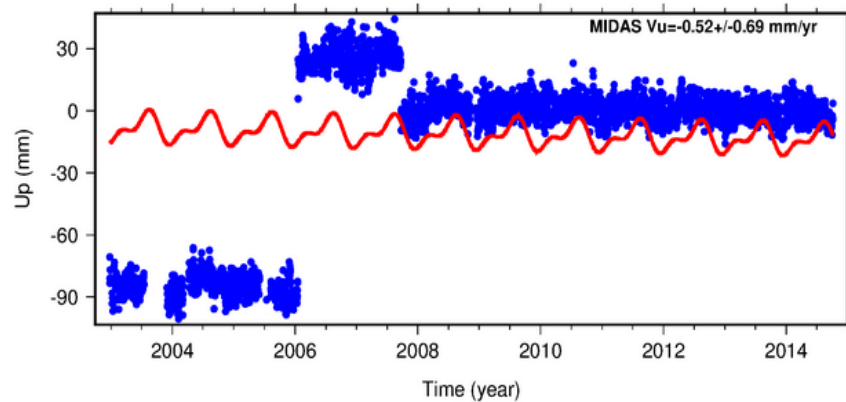
Example: GNSS position time series

Station position time series anatomy

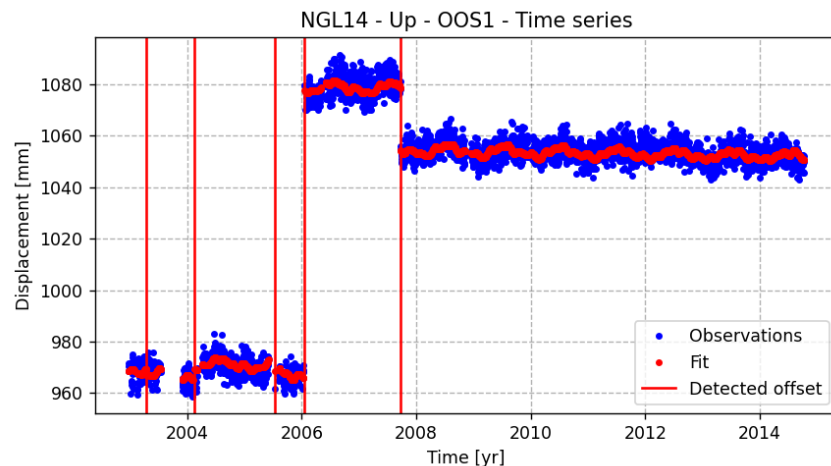


The problem of discontinuities

Time series without offset information



Time series after offset detection



Many offset detection methods

Manual | Ad-hoc | Penalized likelihood |
Hypothesis testing | Bayesian

Gazeaux et al. (2013) – Comparison of 25 methods/solutions

Gazeaux et al. (2015) – Penalized likelihood

Bertin et al. (2017) – Penalized likelihood

Teunissen (2018) – Hypothesis testing

Amiri-Simkooei et al., (2019) – Hypothesis testing

An optimal method?

- Probably not (Teunissen, 2018)
- Ill-posed problem, especially if the noise is not well-known

Can we have good heuristics ?

- Certainly, for some specific applications

Generalized Likelihood Ratio Test (GRLT)

In theory: nested models

Hypotheses

H_0 : **without** additional discontinuity

H_a : **with** additional discontinuity

Linear models

$$H_0 : E\{\underline{\mathbf{y}}\} = \mathbf{A}\mathbf{x}$$

$$H_a : E\{\underline{\mathbf{y}}\} = \mathbf{A}\mathbf{x} + \mathbf{C}_y \nabla$$

Weighted Least-Square Solutions

$$\max_{\mathbf{x}_0 \in \mathbb{R}^n} p_{\underline{\mathbf{y}}}(\mathbf{y}|\mathbf{x}_0),$$

$$\max_{\mathbf{x}_a \in \mathbb{R}^n, \nabla \in \mathbb{R}^q} p_{\underline{\mathbf{y}}}(\mathbf{y}|\mathbf{x}_a, \nabla),$$

Study of a T-statistic

$$\underline{T}_q = -2 \ln(\underline{\text{GLR}}) = \underline{\hat{\mathbf{e}}}_0^T \mathbf{Q}_y^{-1} \underline{\hat{\mathbf{e}}}_0 - \underline{\hat{\mathbf{e}}}_a^T \mathbf{Q}_y^{-1} \underline{\hat{\mathbf{e}}}_a,$$

Reject H_0 if T is *too high*

$$\underline{T}_q = \underline{\hat{\mathbf{e}}}_0^T \mathbf{Q}_y^{-1} \mathbf{C}_y [\mathbf{C}_y^T \mathbf{Q}_y^{-1} \mathbf{P}_A^\perp \mathbf{C}_y]^{-1} \mathbf{C}_y^T \mathbf{Q}_y^{-1} \underline{\hat{\mathbf{e}}}_0,$$

Parsimony is introduced by statistical significance

$$H_0 : \underline{T}_q \sim \chi^2(q, 0).$$

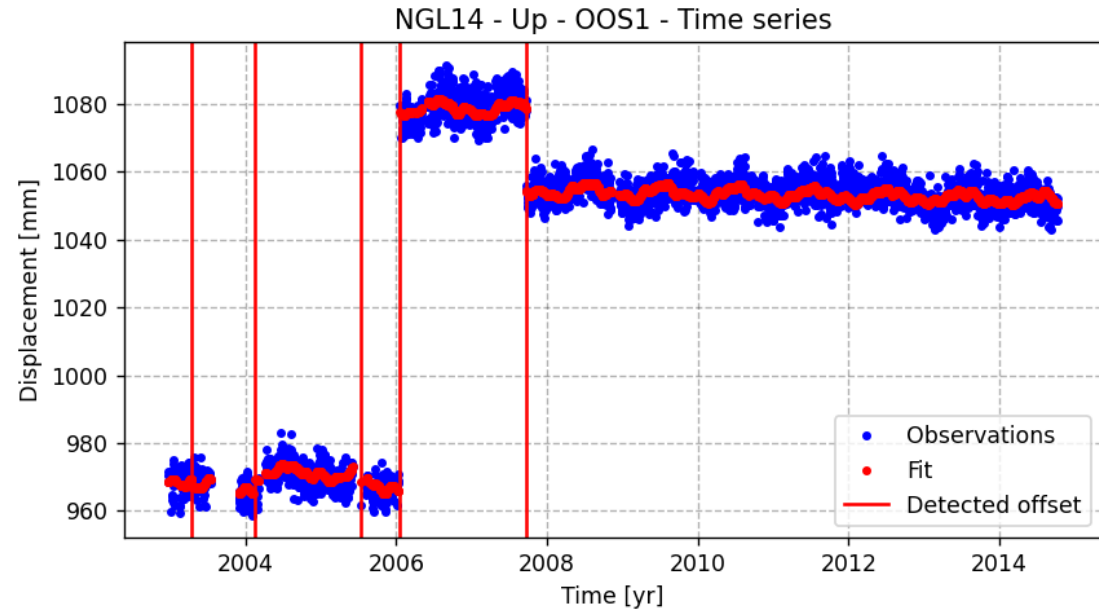
$$\underline{T}_q > \chi_\alpha^2(q, 0)$$

Generalized Likelihood Ratio Test (GRLT)

In practice

GRLT requires 3 main ingredients

1. **Realistic noise model**
2. **Initial trajectory model**
3. **Discontinuity models**



1. Realistic noise model

- Literature
- Dedicated stochastic modelling (beware of estimation biases)

2. Initial trajectory model

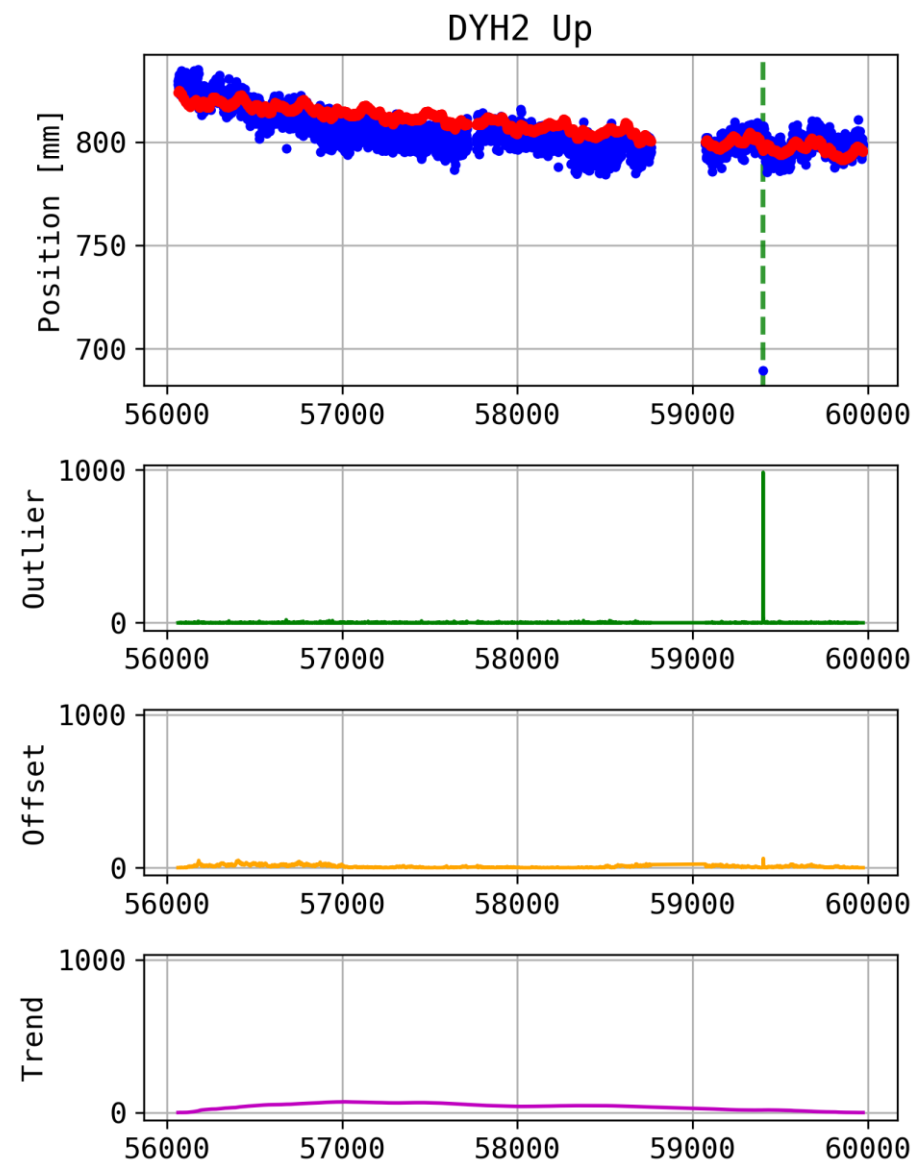
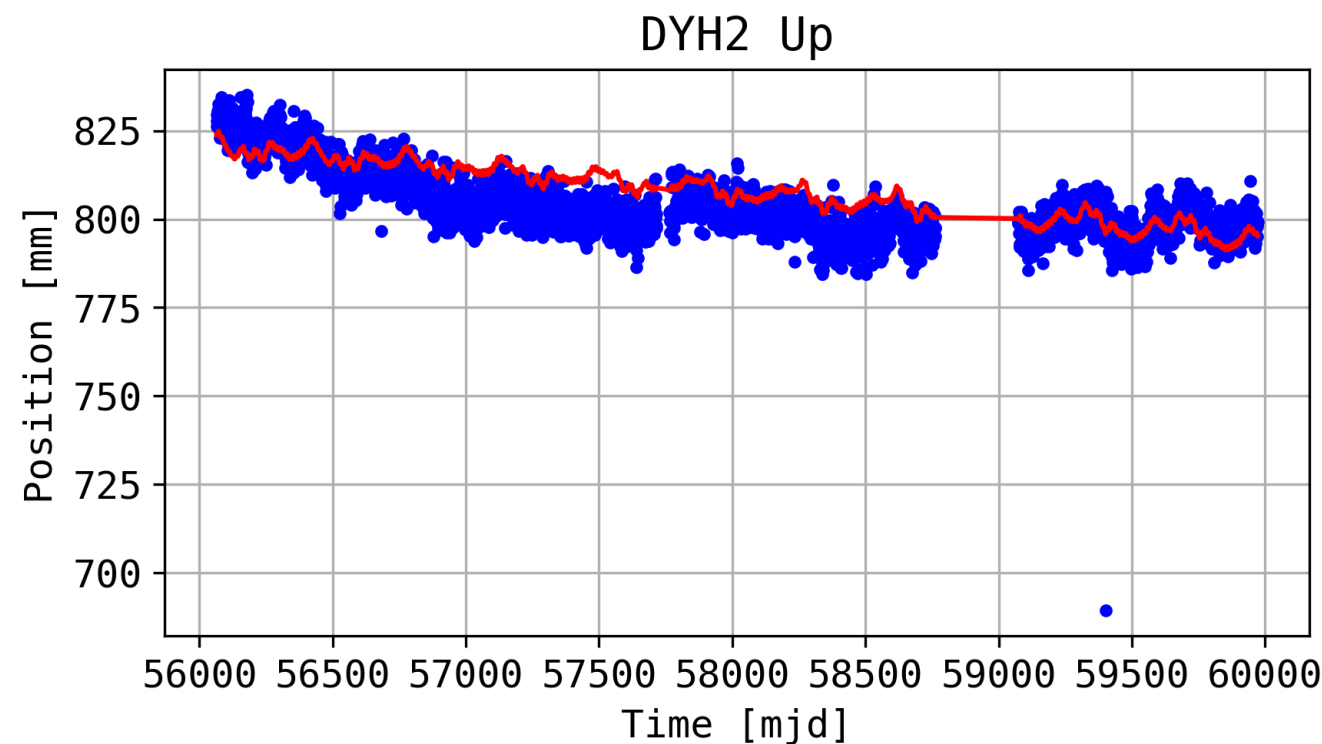
- Literature
- Metadata
- Known events

3. Discontinuity models

- 1 outlier
- 1 offset
- 1 velocity change

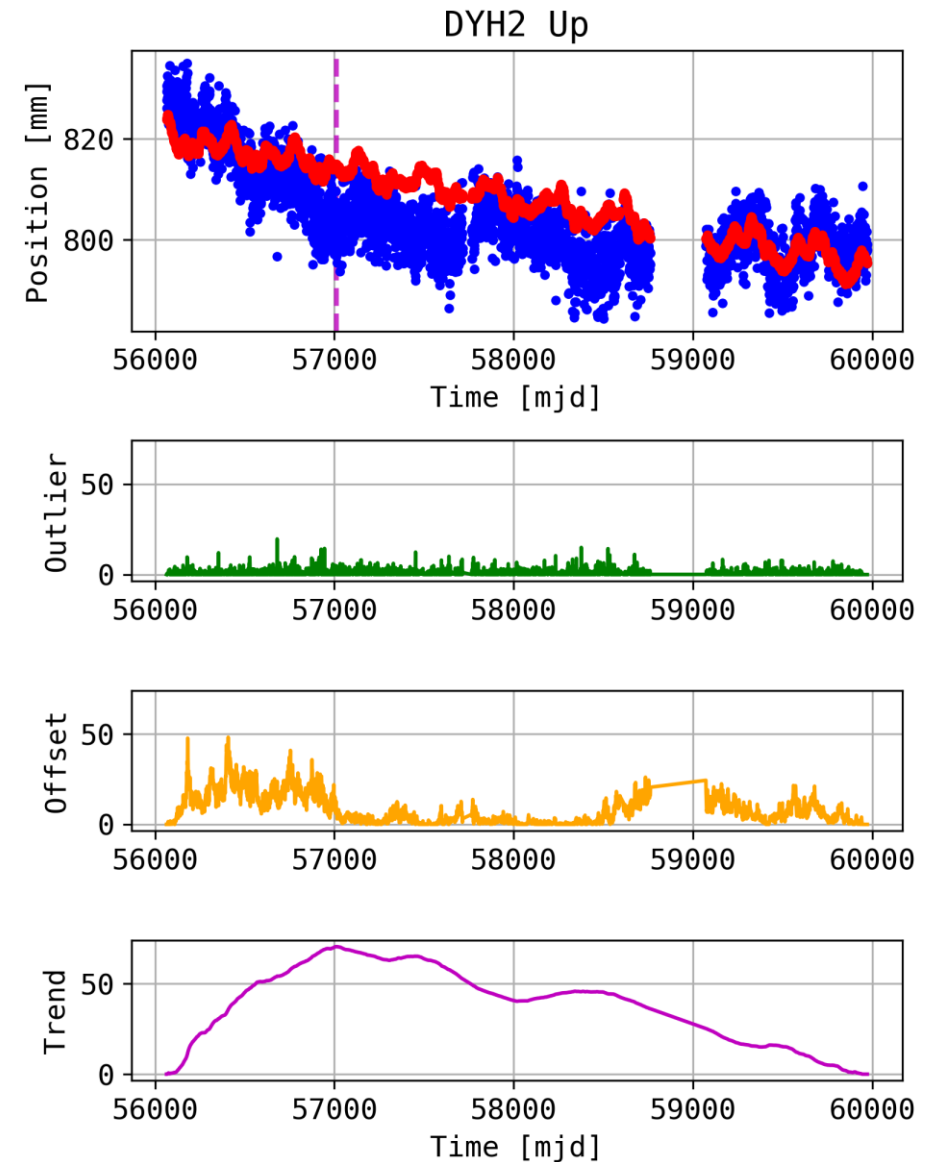
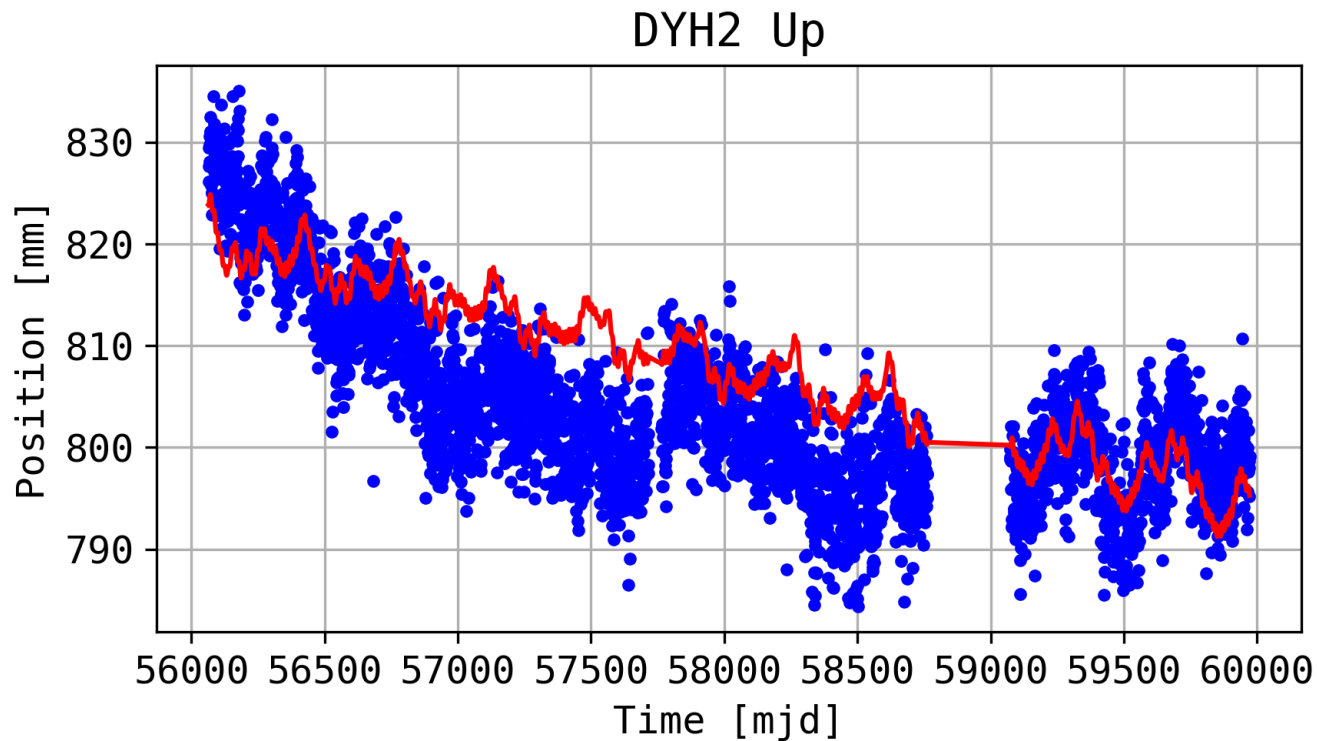
Generalized Likelihood Ratio Test (GRLT) - Example

Initial trajectory model



Generalized Likelihood Ratio Test (GRLT) - Example

Corrected trajectory model



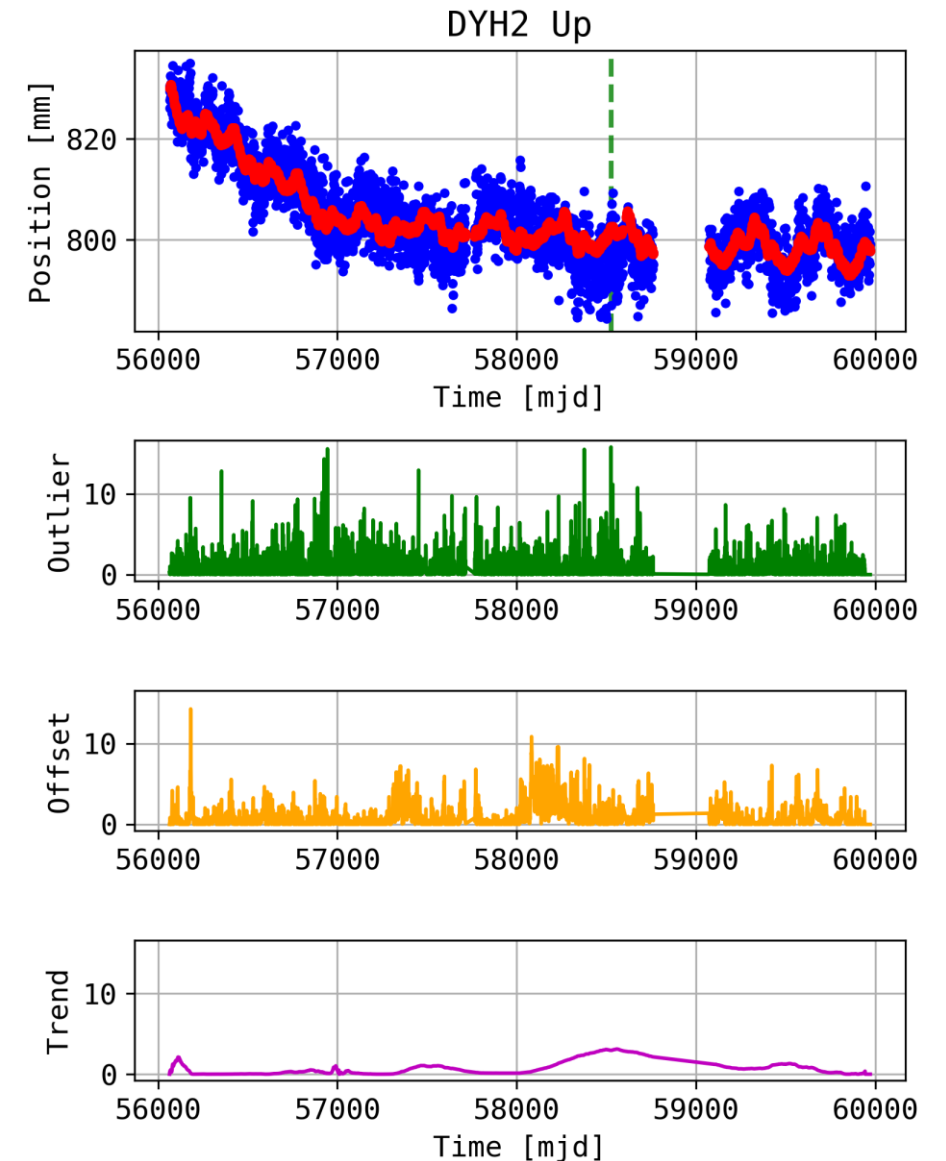
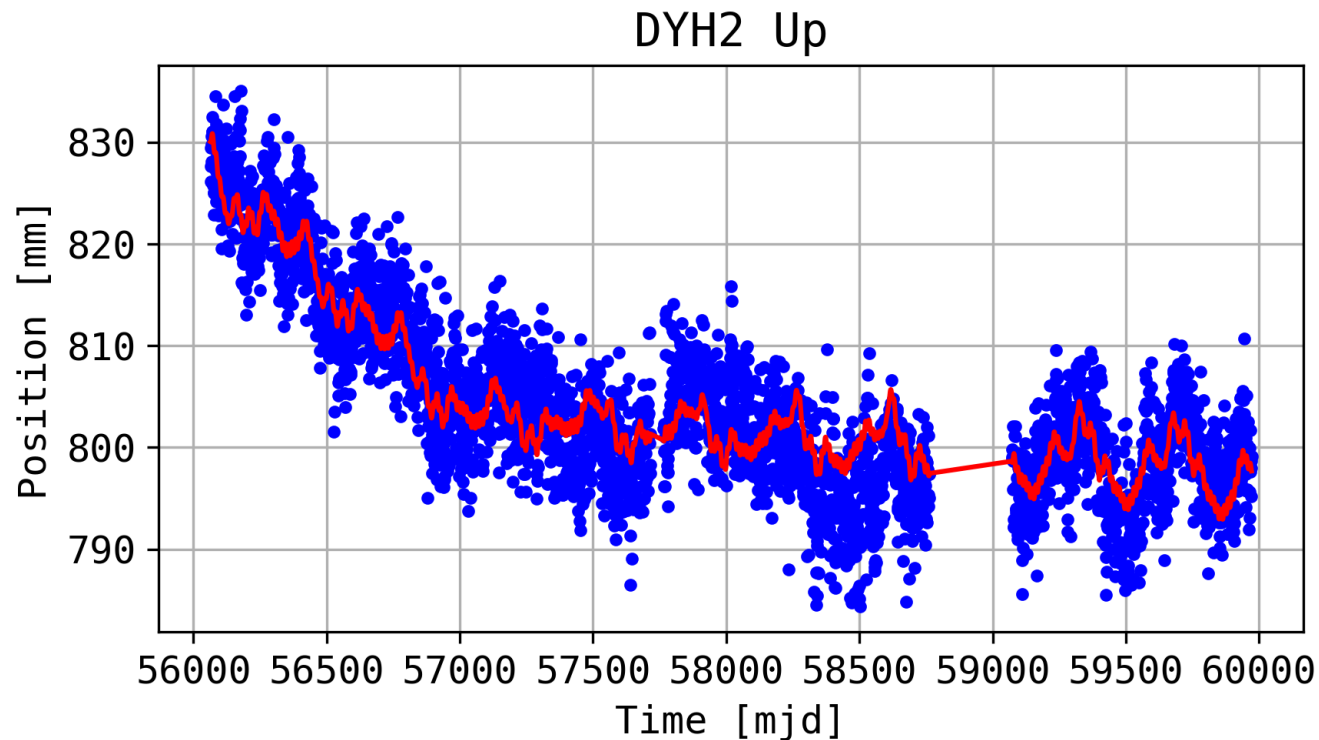
Generalized Likelihood Ratio Test (GRLT) - Example

Final trajectory model

Could be improved with better algorithm than iteration

Can be improved with **multivariate analysis**

Can be improved with **spatial analysis**



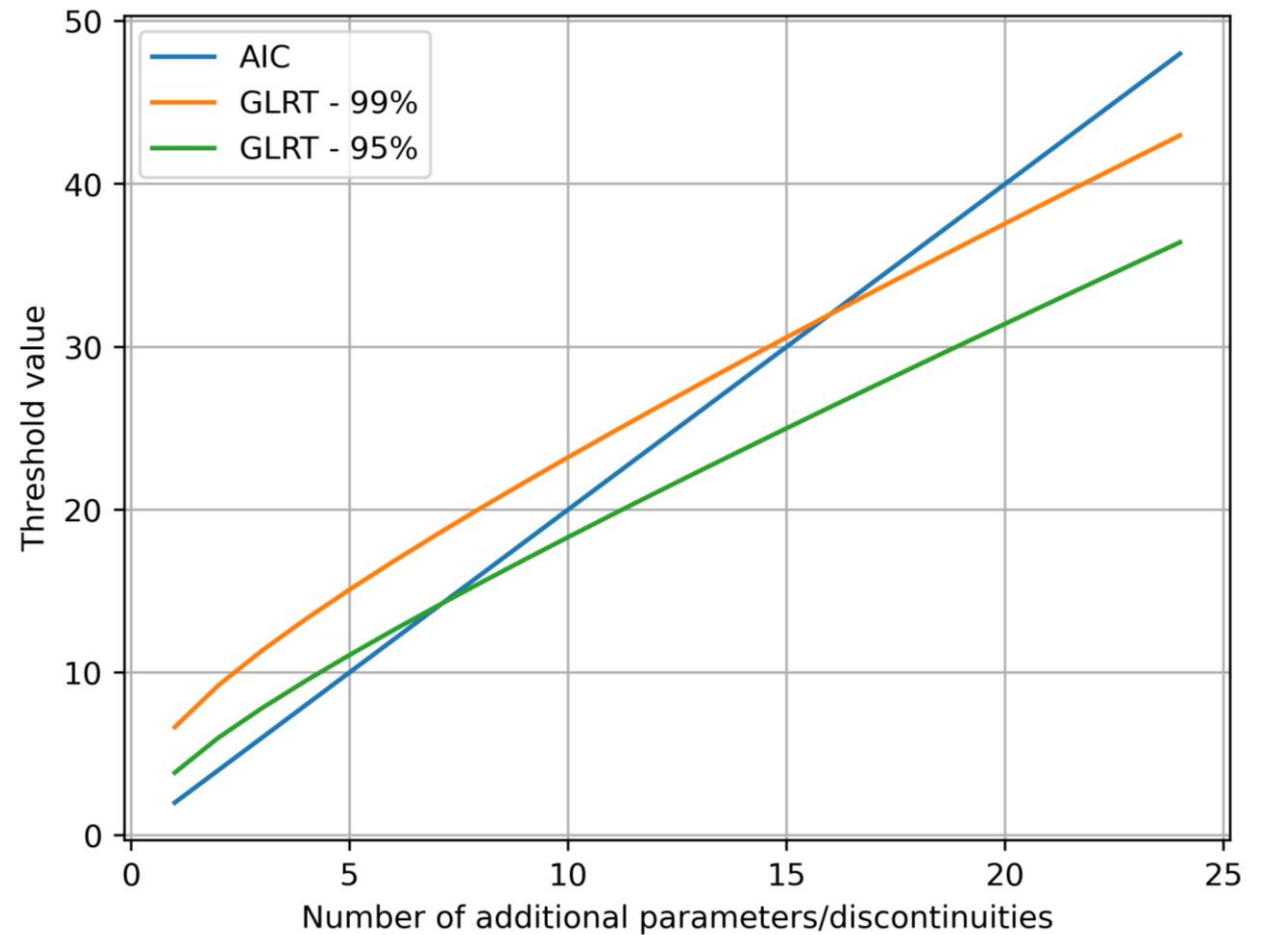
GRLT vs AIC

Generalized Likelihood Ratio Test

$$\underline{T}_q > \chi^2_{\alpha}(q, 0)$$

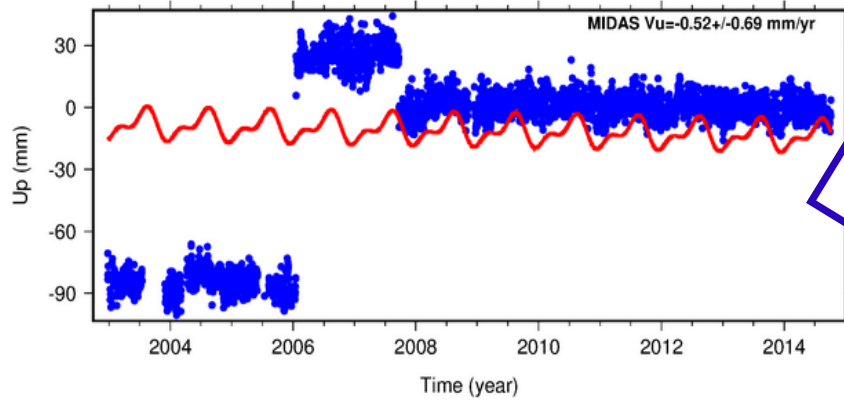
Akaike Information Criterion

$$T_q > 2q$$

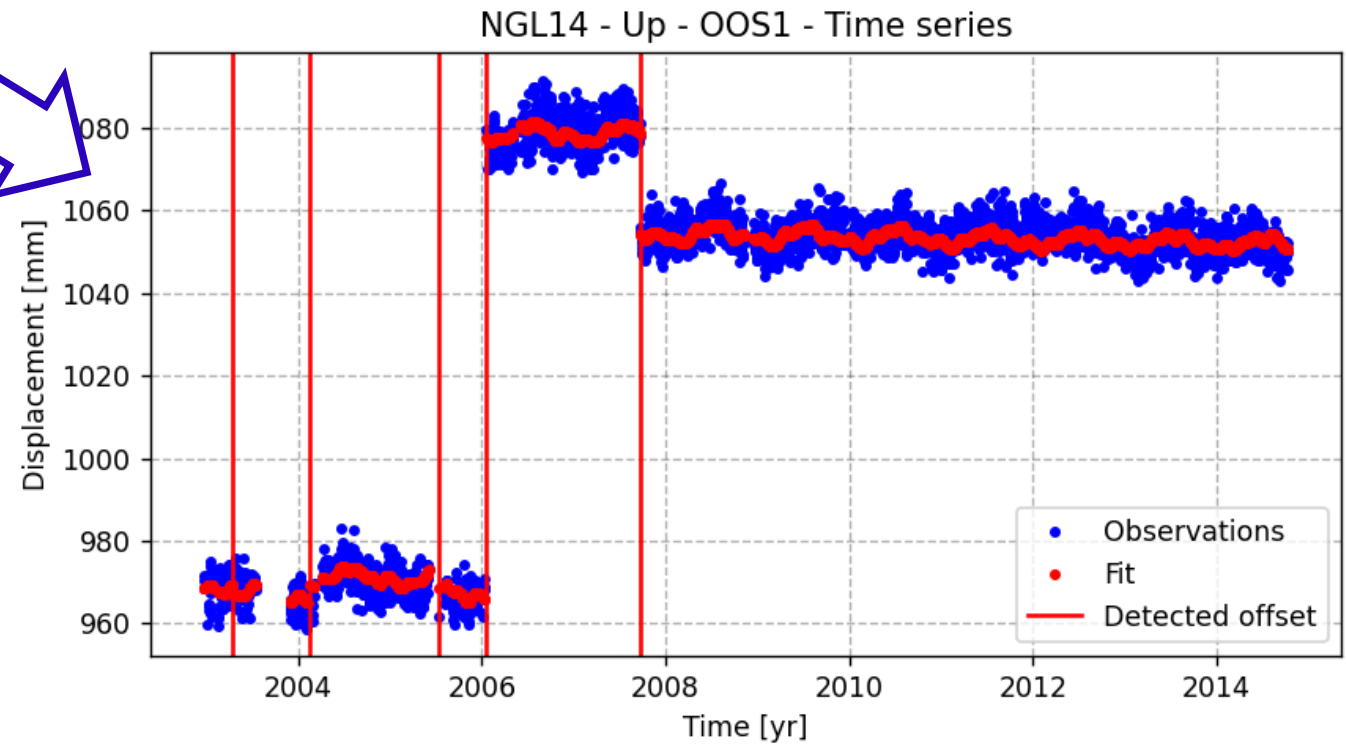


GNSS usage: get a more realistic trajectory model

Starting model

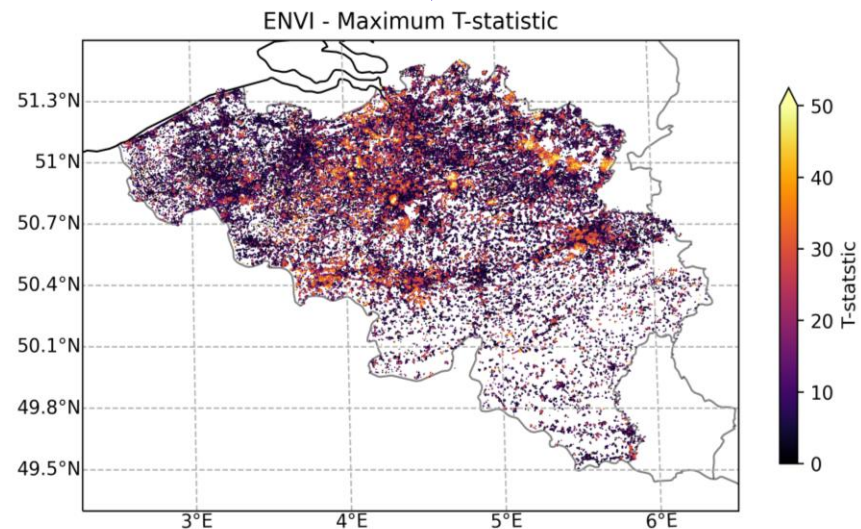
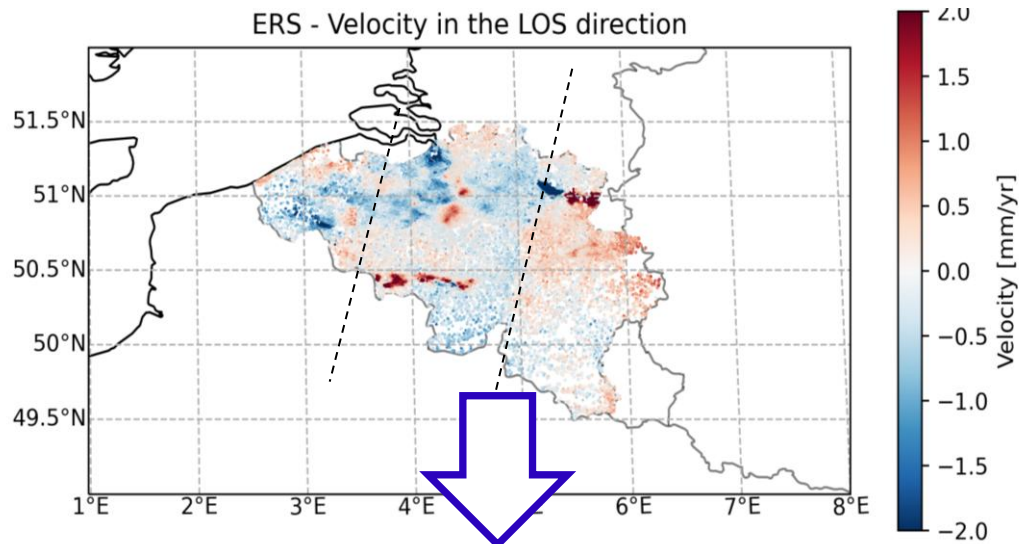


Final model

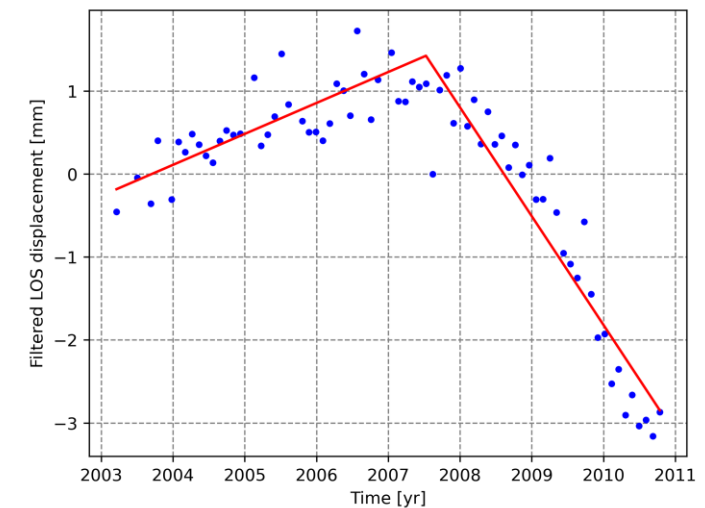
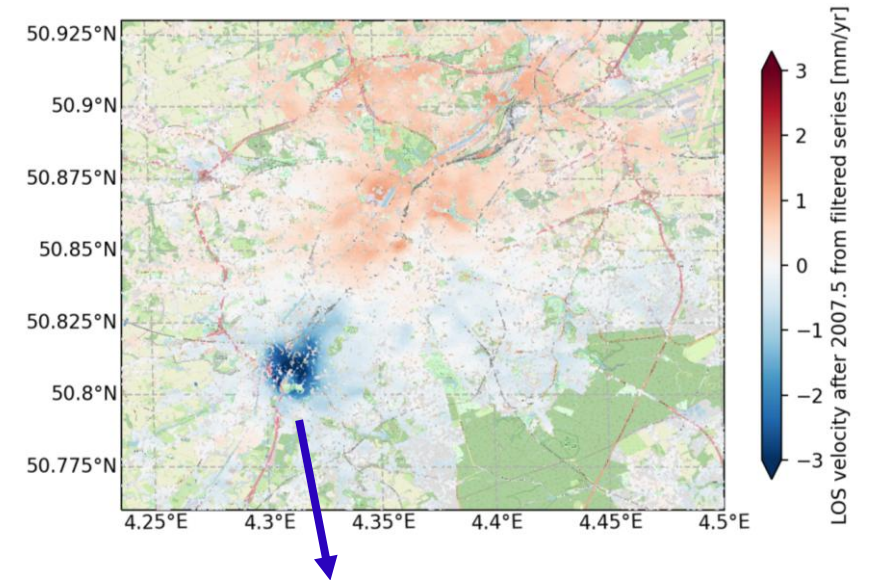


InSAR usage: find areas subject to changes in deformation

ERS (1992-2005)
LOS velocities

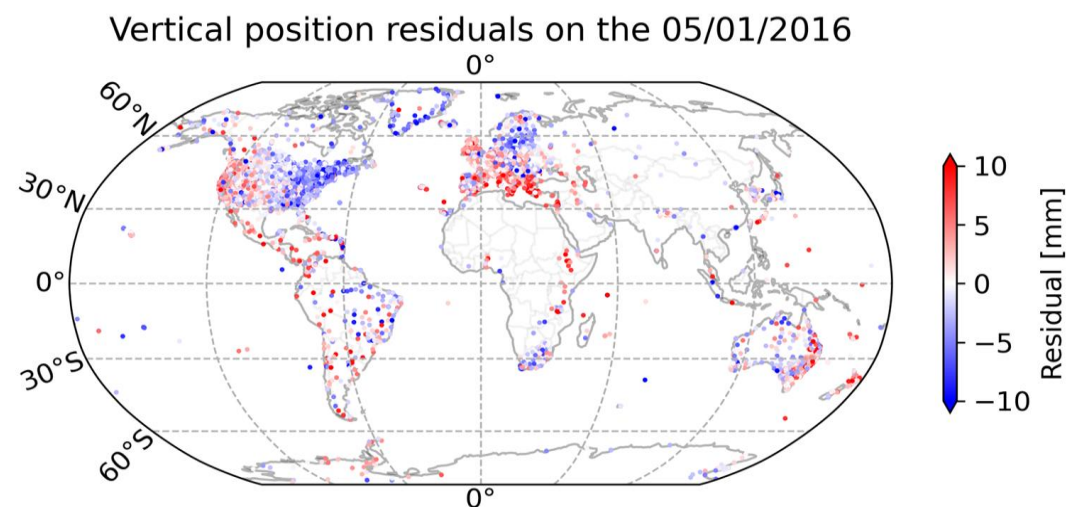


Change in Brussels



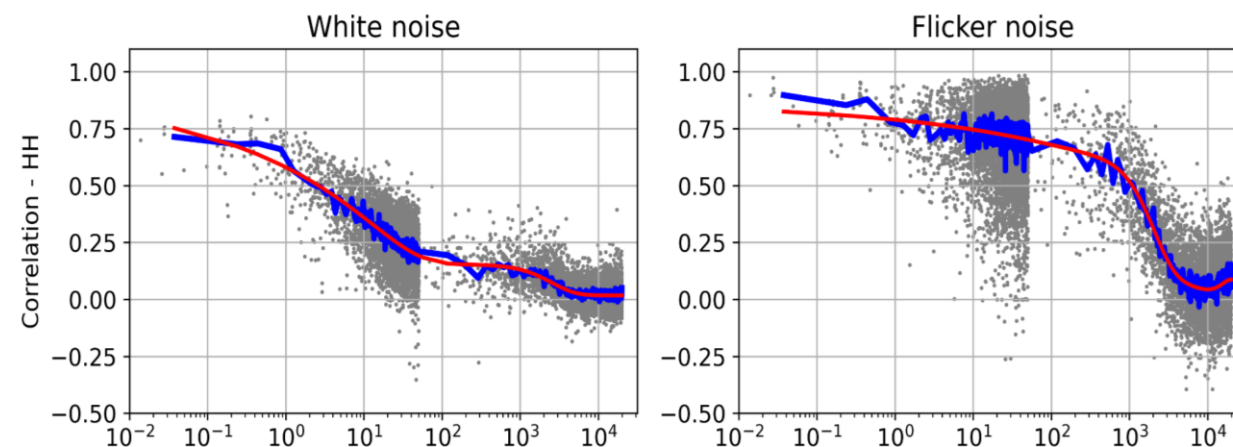
Spatial correlations

Spatial correlations (Wdowinski et al., 1997)



Position time series corrected for loading deformation using the GFZ model (NTAL+NTOL+HYDL+SLEL)

White noise and flicker noise spatial correlations



Diagnostic et modélisation mathématique

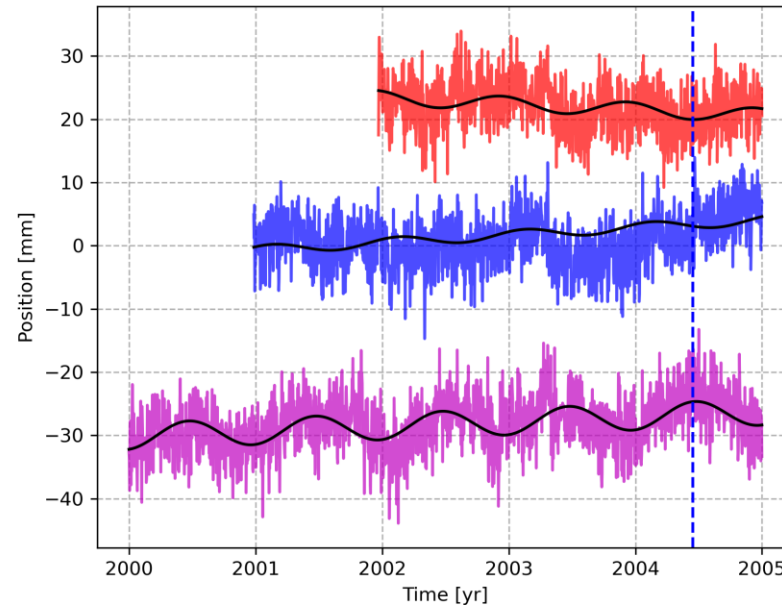
- ✓ Diagnostic (Gobron et al., 2024)
- ✓ Modélisation mathématique (Rebischung & Gobron, 2024)

On the use of spatial correlations for event detection

Simulation:

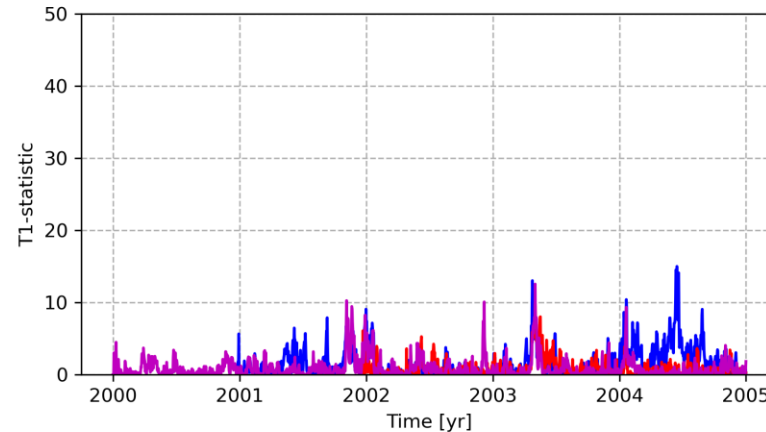
- 3 nearby vertical position time series
- Realistic noise properties
- Flicker noise correlation of 0.8.
- No white noise correlation
- One 6 mm offset in the middle time series

Spatially blind detection



Two offset detection tests:

- Spatially blind detection
- Spatially aware detection



On the use of spatial correlations for event detection

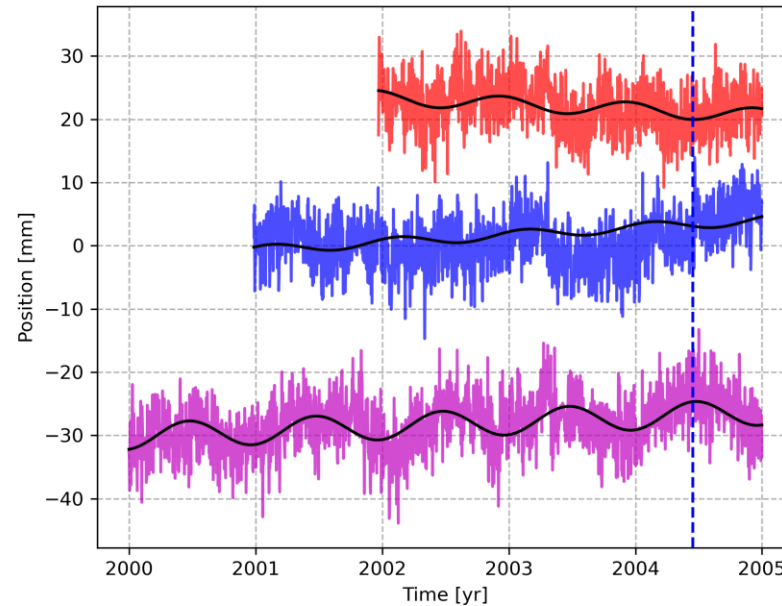
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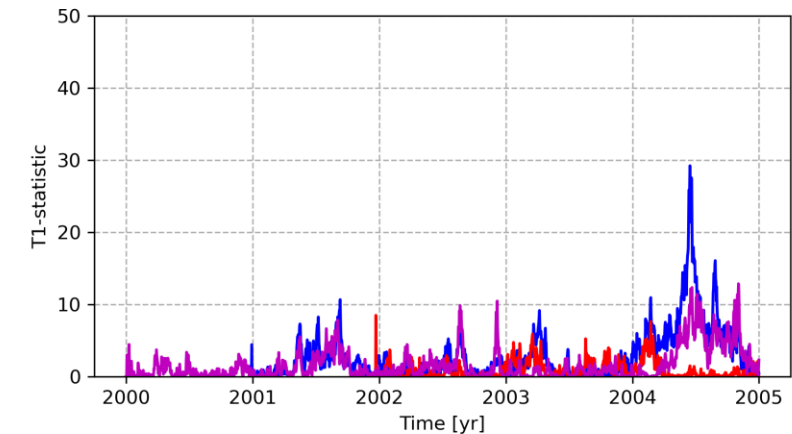
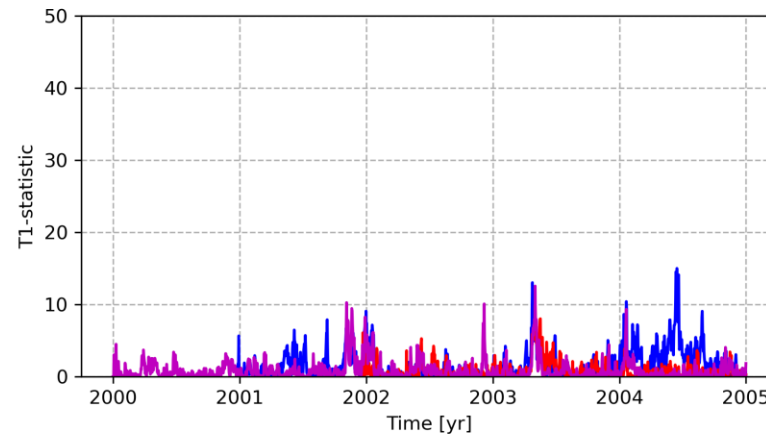
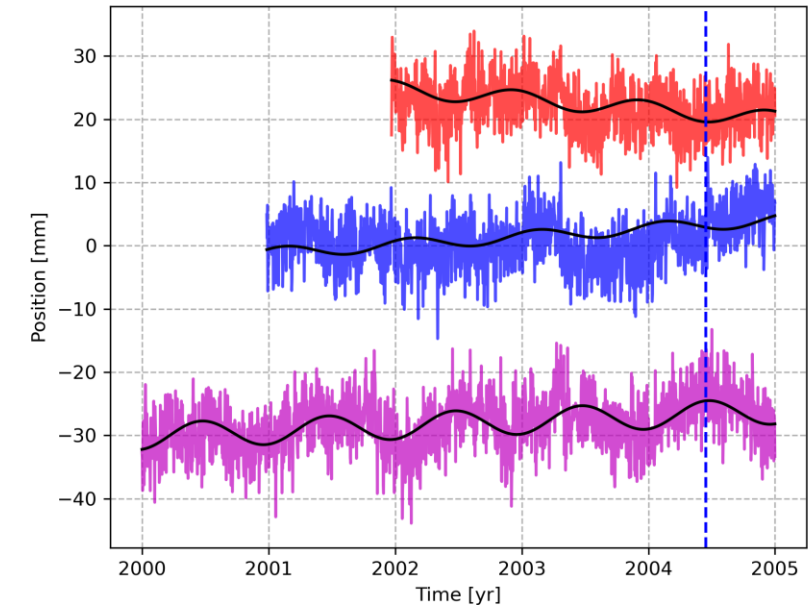
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- Spatially blind detection
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Spatially blind detection

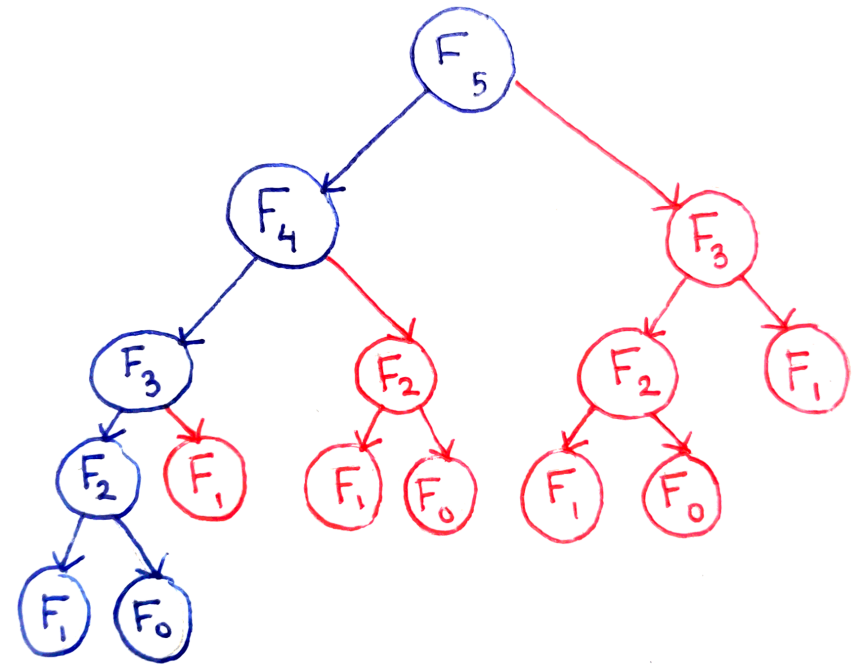


Spatially aware detection



Other possible long-term improvements

- **Better exploration of the tree of possibilities**
 - Explore multiple discontinuities at once (but cumbersome).
 - AI algorithms for change detection known « sparse dictionary learning » using deep temporal dictionary learning or deep symbolic regression methods?
 - **More realistic trajectory models?**
 - **Faster modelling step?**



Conclusion

Take home message:

- There is probably **no perfect change detection** method
- **Generalized Likelihood Ratio Test (GLRT)** nonetheless provides a flexible framework for change detection.
- However, it requires **realistic noise models**
- We are improving this using **spatial correlation** modelling
- **AI approaches** could also lead to **faster** and more **robust modelling** strategies in the long run

